Create a Factorial API with a CSHTML Page and Test It Using Postman: The application will have a CSHTML page with a textbox to enter a number, a Calculate Factorial button, an ASP.NET Core API to calculate the factorial, the result displayed on the CSHTML page, and the same API tested separately using Postman.

Part A — Create the Factorial API

1. Create the ASP.NET Core MVC Project: Visual Studio Create a new project ASP.NET Core Web App (Model-View-Controller) Next, **Enter Project Name:** FactorialAPI.

2. Create the API Controller

Select the following options horizontally: Solution Explorer Controllers Right-click Add Controller API Controller – Empty. **Name the controller:** FactorialController.cs. Click Add.

3. Write the Factorial API Code

```
using Microsoft.AspNetCore.Mvc;

namespace FactAPI.Controllers
{
    public class FactorialController : Controller
    {
        // Displays the CSHTML page
        public IActionResult Index()
        {
            return View();
        }

        // API endpoint
        [HttpGet("/api/factorial/{number}")]
        public IActionResult CalculateFactorial(int number)
        {
            if (number < 0)
                return BadRequest("Factorial is not defined for negative numbers.");

            long factorial = 1;

            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }

            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}
```

4. Understand the API URL

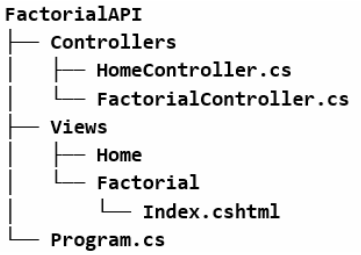
https://localhost:7175/api/factorial/5

Part B — Create the CSHTML Page

5. Create the View

Views Right-click Add New Folder Factorial Right-click Factorial Add View Razor View - Empty Name the view: Index.cshtml. Click Add.

6. Add the CSHTML Code: Open Views Factorial Index.cshtml and add:




```
<div>
  <label>Enter a number: </label>
  <input type="number" id="number" />
  <button onclick="CalculateFactorial()">Calculate Factorial</button>
</div>
<br />
<div>
  <label>Result:</label>
  <input type="text" id="result" readonly />
</div>

<script>
    function CalculateFactorial()
    {
    var number = document.getElementById("number").value;

    fetch('/api/factorial/' + number)
      .then(response => {
        if (!response.ok)
        {
          throw new Error("Invalid Number");
        }

        return response.json();
      })
      .then(data => {
        document.getElementById("result").value = data.factorial;
      })
      .catch(error => {
        alert(error.message);
      });
    }
</script>
```

7. Add a Link to the Factorial Page

Open Views Shared _Layout.cshtml.

Find the navigation section and add:

```
<li class="nav-item">
  <a class="nav-link text-dark" asp-controller="Factorial" asp-action="Index">
    Factorial Calculator
  </a>
</li>
```

The navigation bar will then contain a Factorial Calculator link.

9. Run the Application: For example: https://localhost:7175

10. Open the Factorial Page with input controls: Enter: https://localhost:7175/Factorial

11. Enter a Number: Enter 5 and click Calculate Factorial. The CSHTML page calls GET /api/factorial/5.

Part C — Test the API Using Postman: https://localhost:7175/api/factorial/5

Output:

Postman should return:

```
{
  "number": 5,
  "factorial": 120
}
```

This confirms that the API is working correctly.